

Know the foundations

Notes, formulas, methods and memory warnings

YOUR AIM TODAY

Relearn quadratics, functions, coordinate geometry and circles from the beginning in one organised booklet.

YOUR RULE

Read actively. Cover each worked example, reproduce it, then say why each step is valid. Use the practice booklet only after finishing these notes.

Student: Rohan

Start time: _____ Finish time: _____

Date: _____

What you must know

These four foundations support almost every later P1 topic.

Topic	You must be able to
Quadratics	Expand; factorise; complete the square; solve; sketch; use the discriminant; solve inequalities and parameter conditions
Functions	Use notation; substitute; find composites and inverses; understand domain, range and one-to-one functions
Coordinate geometry	Find gradient, midpoint and distance; form line equations; use parallel and perpendicular gradients
Circles	Read and construct circle equations; complete squares; verify points; find tangent equations and intersections

THE UNIVERSAL HABIT

Do not begin with arithmetic. Name the method, write its formula or structure, then substitute values.

Symbols you will see

Symbol	Meaning
D	Discriminant $b^2 - 4ac$
$f(x)$	Output from function f for input x
$fg(x)$	$f(g(x))$; g happens first
$f^{-1}(x)$	Inverse function; not $1/f(x)$
m	Gradient of a straight line
r	Radius of a circle

Algebra and completing the square

A quadratic has highest power x^2 .

Form	Example	Best use
Expanded	x^2-6x+5	Read coefficients; discriminate
Factorised	$(x-1)(x-5)$	Read roots
Completed square	$(x-3)^2-4$	Read turning point and range

EXPAND BRACKETS

Multiply every term by every term. $(x-4)(2x+3)=2x^2+3x-8x-12=2x^2-5x-12$.

Completing the square when the x^2 coefficient is 1

- For x^2+bx+c , halve b inside the bracket.
- Square the bracket, then compensate for the extra constant.

WORKED EXAMPLE

$x^2-8x+7=(x-4)^2-16+7=(x-4)^2-9$. Turning point: $(4,-9)$.

When the x^2 coefficient is not 1

WORKED EXAMPLE

$2x^2+12x+5=2(x^2+6x)+5=2[(x+3)^2-9]+5=2(x+3)^2-13$.

REMEMBER

The turning point of $y=a(x-h)^2+k$ is (h,k) . The sign inside the bracket reverses.

Solving, discriminants and graphs

Choose factorising, completed square or the quadratic formula.

Method	When useful	Key line
Factorise	Integer/rational roots visible	Set each factor to zero
Complete square	Completed form already known	Square-root gives plus and minus
Formula	Does not factorise easily	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

WORKED SOLUTION

Solve $x^2 - 6x + 2 = 0$: $(x-3)^2 - 7 = 0$, so $(x-3)^2 = 7$, $x-3 = \pm\sqrt{7}$, therefore $x = 3 \pm \sqrt{7}$.

The discriminant

$D = b^2 - 4ac$	Roots	Graph
$D > 0$	Two distinct real roots	Crosses x-axis twice
$D = 0$	One repeated real root	Touches x-axis
$D < 0$	No real roots	Does not meet x-axis

PARAMETER EXAMPLE

$x^2 + 6x + k = 0$ has a repeated root: $D = 36 - 4k = 0$, so $k = 9$.

IMPORTANT

In $ax^2 + bx + c$, a , b and c are numbers. Include their signs. If the equation is not equal to zero, rearrange it first.

Inequalities and parameter conditions

Roots divide the number line into sign regions.

Method for a quadratic inequality

- Move everything to one side so the other side is zero.
- Find the boundary roots.
- Decide where the parabola is positive or negative.
- Include roots for $\geq$ or $\leq$; exclude them for $>$ or $<$.

BETWEEN THE ROOTS

Solve $x^2 - 5x + 6 < 0$. Factor: $(x-2)(x-3) < 0$. The parabola opens upward, so it is below the axis between its roots: $2 < x < 3$.

OUTSIDE THE ROOTS

Solve $x^2 - 5x + 6 \geq 0$. The same parabola is on or above the axis outside its roots: $x \leq 2$ or $x \geq 3$.

Parameters and numbers of roots

- Two real roots: $D > 0$.
- Repeated root: $D = 0$.
- No real roots: $D < 0$.

WORKED PARAMETER CONDITION

$x^2 + px + 16 = 0$ has no real roots: $p^2 - 64 < 0$, so $p^2 < 64$ and therefore $-8 < p < 8$.

COMMON TRAP

From $p^2 < 64$, do not write only $p < 8$. The lower boundary -8 is equally important.

Machines, composites and inverses

A function gives one output for each allowed input.

Idea	Method	Example
Substitution	Replace every x ; use brackets	$f(x)=2x-3$, $f(-4)=-11$
Composite	Inside function first	$fg(x)=f(g(x))$
Inverse	Swap x and y ; rearrange	f^{-1} undoes f
Domain	Allowed inputs	Exclude division by zero
Range	Outputs produced	x^2+1 has range $y \geq 1$

COMPOSITE EXAMPLE

Let $f(x)=2x+1$ and $g(x)=x^2-3$. Then $fg(x)=f(x^2-3)=2(x^2-3)+1=2x^2-5$. But $gf(x)=g(2x+1)=(2x+1)^2-3=4x^2+4x-2$.

INVERSE EXAMPLE

For $f(x)=3x-5$: $y=3x-5$; swap to $x=3y-5$; rearrange $y=(x+5)/3$. Therefore $f^{-1}(x)=(x+5)/3$.

One-to-one warning

- A function needs to be one-to-one before it has an inverse function.
- $g(x)=x^2$ is not one-to-one on all real numbers because $g(2)=g(-2)$.
- Restricting the domain to $x \geq 0$ makes the inverse $\sqrt{x}$.

REMEMBER

$f^{-1}(x)$ means inverse function, not reciprocal. fg and gf are usually different.

Points and straight lines

Gradient controls direction; one point fixes position.

Skill	Formula
Gradient	$m = (y_2 - y_1) / (x_2 - x_1)$
Midpoint	$((x_1 + x_2) / 2, (y_1 + y_2) / 2)$
Distance	$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Line through point	$y - y_1 = m(x - x_1)$
Parallel	Equal gradients
Perpendicular	$m_1 * m_2 = -1$; negative reciprocals

ONE COMPLETE EXAMPLE

$A = (1, 2)$, $B = (7, 10)$. Gradient $= (10 - 2) / (7 - 1) = 4/3$. Midpoint $= (4, 6)$. Distance $= \sqrt{6^2 + 8^2} = 10$. Line: $y - 2 = (4/3)(x - 1)$.

PERPENDICULAR BISECTOR

It needs two facts: it passes through the midpoint, and its gradient is the negative reciprocal. For the example, $m = -3/4$ and $y - 6 = (-3/4)(x - 4)$.

Checks

- Substitute a known point into your final line equation.
- Distance must be non-negative.
- A midpoint should lie between the endpoints.
- Vertical lines have undefined gradient and equation $x = \text{constant}$.

COMMON TRAP

For gradient, subtract coordinates in the same order on top and bottom.

Centre, radius, points and tangents

Circle form converts algebra into geometry.

STANDARD FORM

$(x-a)^2+(y-b)^2=r^2$ has centre (a,b) and radius r . Signs inside brackets reverse when reading the centre.

Complete two squares

WORKED EXAMPLE

$x^2+y^2-4x+6y-12=0$ becomes $(x-2)^2-4+(y+3)^2-9-12=0$, so $(x-2)^2+(y+3)^2=25$. Centre $(2,-3)$, radius 5.

Point on a circle

- Substitute the point into the circle equation.
- It lies on the circle only if left side equals r^2 .

Tangent at a point

- Find the centre.
- Find the radius gradient from centre to contact point.
- Take the negative reciprocal for tangent gradient.
- Use the contact point in $y-y_1=m(x-x_1)$.

TANGENT EXAMPLE

Centre $(2,-3)$, contact $P=(5,1)$: radius $m=4/3$, tangent $m=-3/4$. Hence $y-1=(-3/4)(x-5)$, or $3x+4y=19$.

INTERSECTIONS

Substitute the line into the circle. Solve the resulting quadratic or squared equation, then report complete coordinate pairs.

Important points to remember

Say each rule aloud before the practice booklet.

Rule	I can explain it
Completing square: halve the x coefficient inside the bracket.	
A square-root equation usually gives plus and minus roots.	
Discriminant: positive two roots, zero repeated, negative none.	
An upward quadratic is negative between distinct roots and positive outside.	
For a composite, perform the function closest to x first.	
An inverse is not a reciprocal; one-to-one matters.	
Gradient subtraction order must match in numerator and denominator.	
Perpendicular gradients are negative reciprocals.	
Circle centre signs reverse inside brackets.	
A tangent is perpendicular to the radius at the contact point.	

READY TEST

Close this booklet. On blank paper, reproduce the quadratic formula, discriminant rule, composite order, inverse method, five coordinate formulas/rules and circle standard form.